



NOVA PAY

Machine-Learning Fraud Detection System

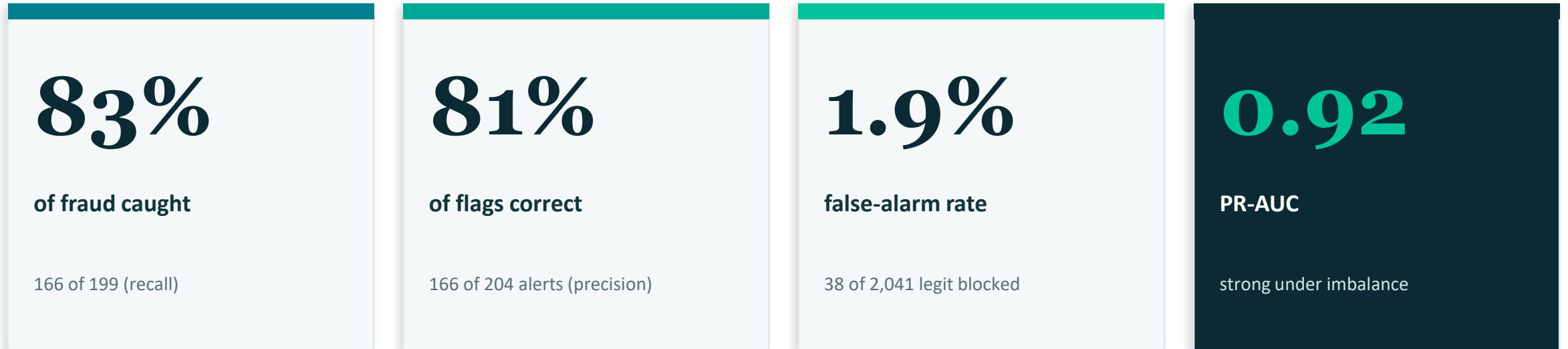
Validated results: an adaptive, explainable system that outperforms rules

Executive Briefing · Overview · Insights · Cost · Recommendations

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What the validated model delivers

Measured on a held-out test set of 2,240 transactions (8.9% fraud) the model never trained on.



In plain terms: the system catches roughly five of every six fraudulent transfers while wrongly stopping under 2% of legitimate ones — a balance hand-written rules cannot reach, and one that can be dialled to appetite.

The problem we are solving

Digital transfer fraud is rare, fast-moving and costly. The current rule-based engine cannot keep pace.



8.9%

of transactions are fraudulent — severe imbalance that rules handle poorly



Static

rules are hand-written, quickly become out-dated, and miss new fraud patterns



Reactive

blunt thresholds drive false alarms — blocking good customers, overloading analysts

The objective: catch more fraud, raise fewer false alarms, explain every decision, and adapt automatically as fraud evolves.

An end-to-end, eight-stage solution

Each stage is engineered for robustness and hands off cleanly to the next — from raw data to a monitored, self-renewing model.



1

Collect & Profile

Baseline fraud prevalence, missingness



2

Clean & Prepare

KNN imputation, outlier capping



3

Engineer Features

Velocity, mismatch, device, behaviour



4

Explore (EDA)

Patterns by channel, geography, time



5

Model & Tune

LogReg → RF → XGBoost / LightGBM



6

Validate & Explain

Holdout, SHAP transaction reasoning



7

Deploy

FastAPI /score, Docker container

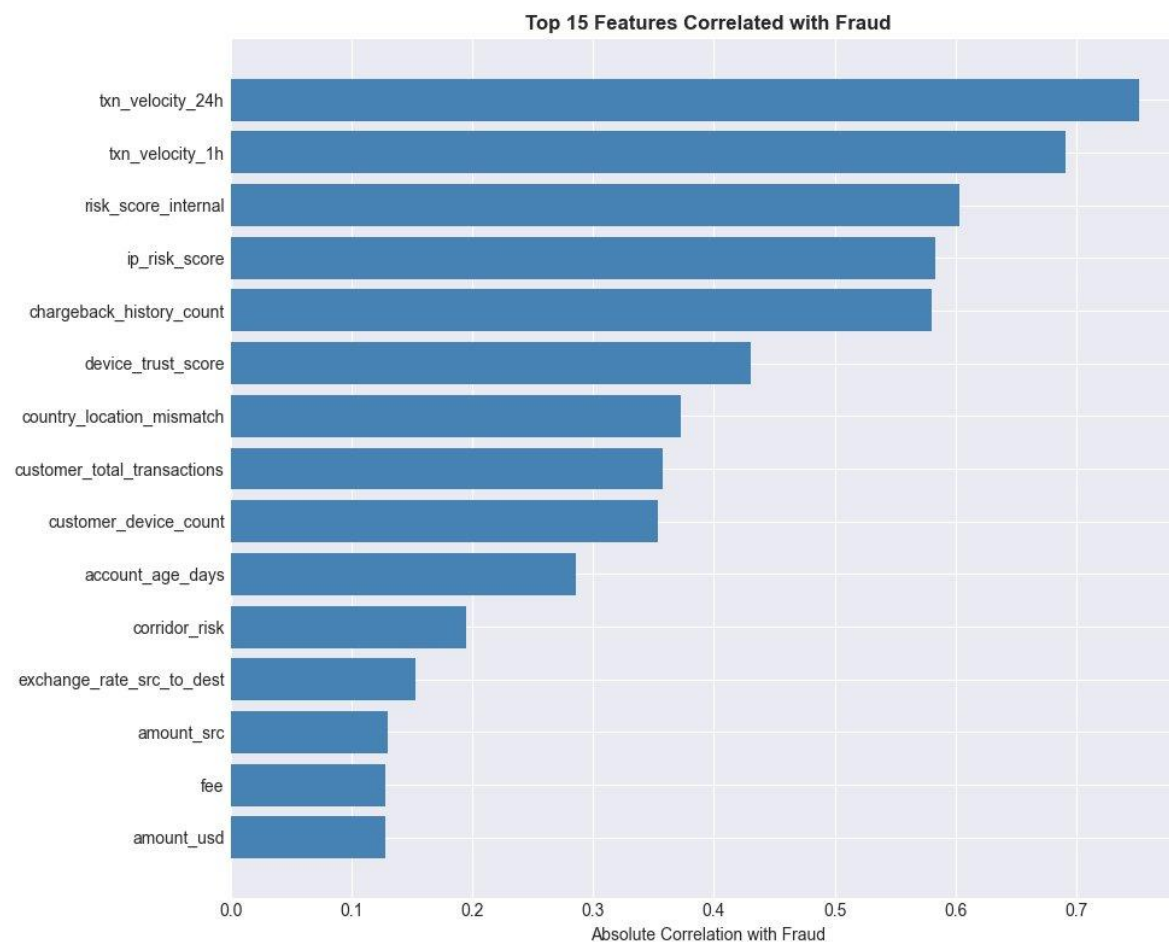


8

Monitor & Improve

Drift detection, quarterly retraining

What actually drives fraud



The signal is behavioural

Transaction velocity — how fast a customer transacts (1h & 24h) — is the strongest fraud signal in the data.

Risk & history scores (internal, IP, chargeback history) and **device trust** follow close behind.

Why it matters: these are behaviour features engineered from data the rules engine never combines — exactly what we kept when we set raw identifiers aside.

Where fraud concentrates: location mismatch

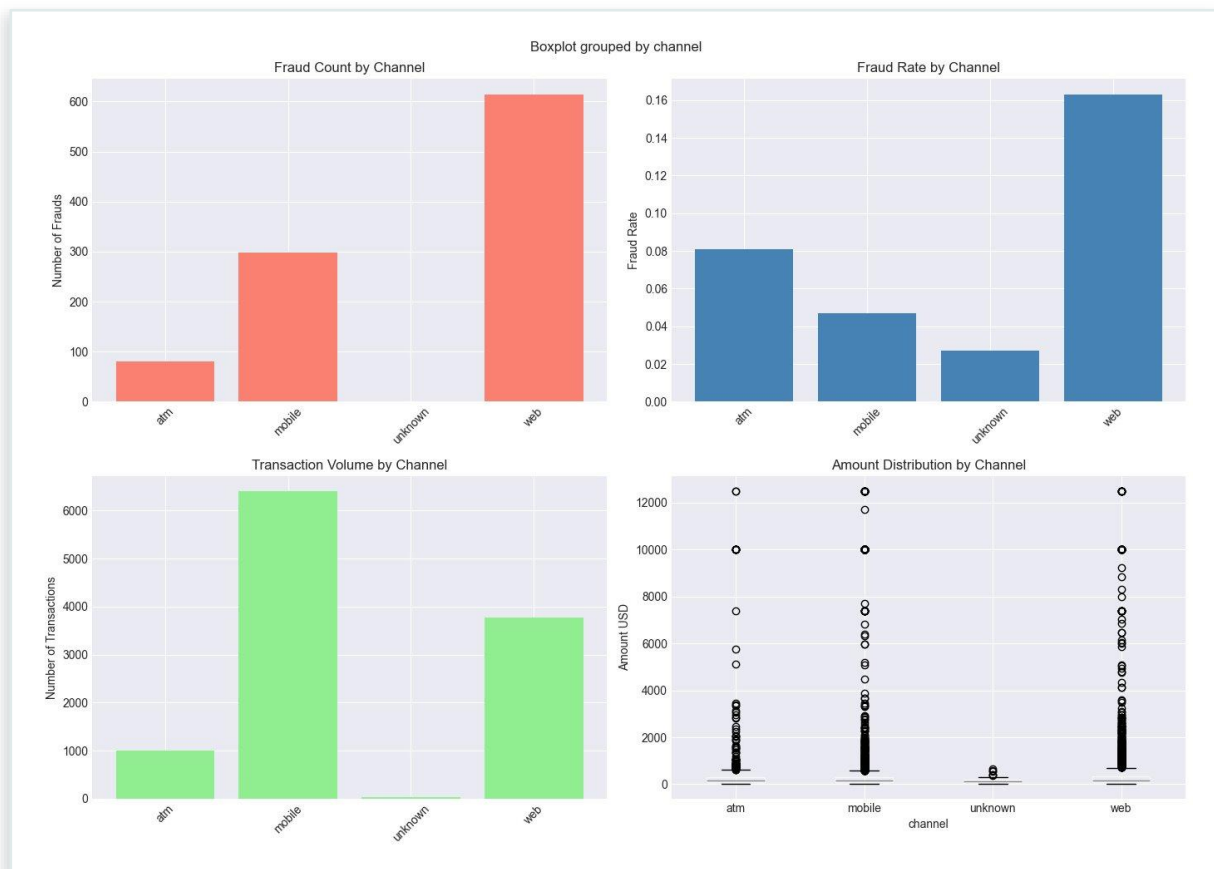


~8× higher

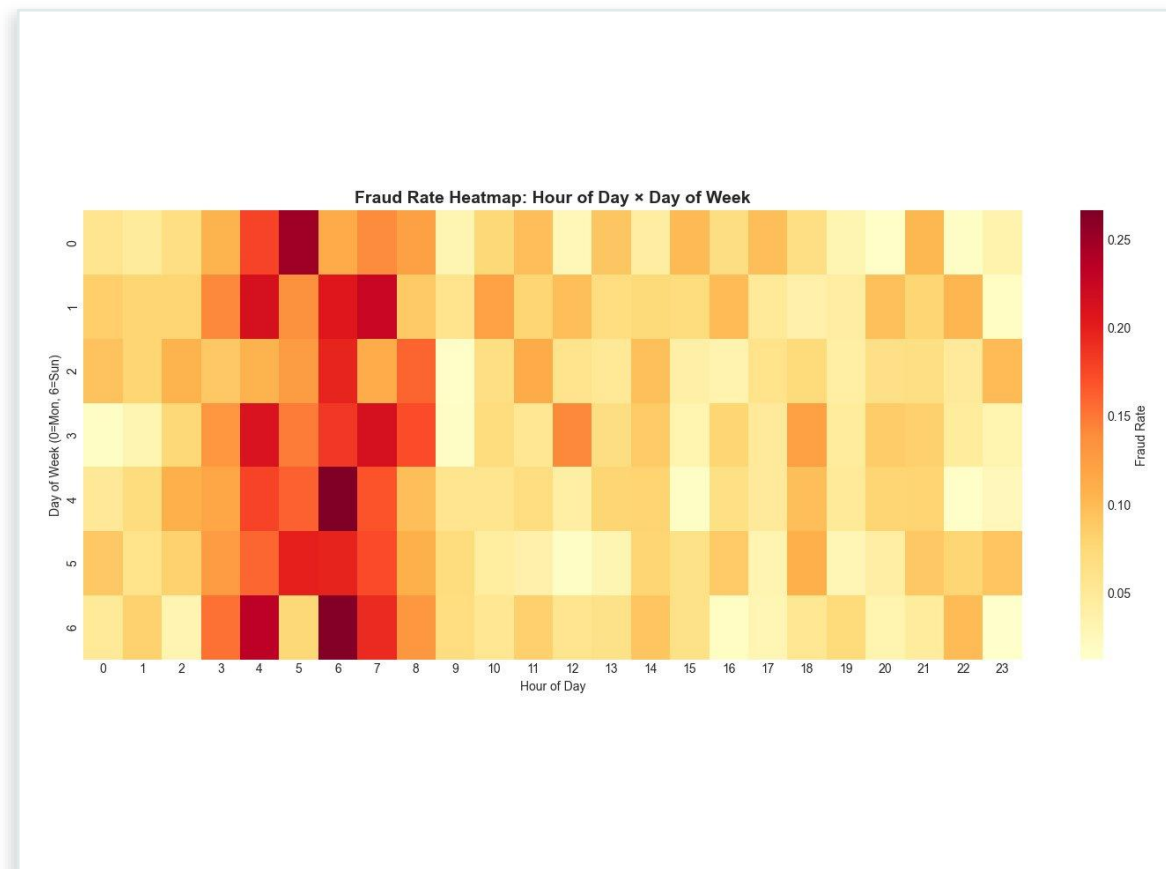
Transactions where customer country and location **do not match** show a fraud rate near **30%**, versus ~3.5% when they agree.

Volume ≠ rate: the US leads on fraud count, but Canada and the UK carry the highest fraud rates — a nuance flat rules miss.

When and where: channel & timing patterns



Web channel: highest fraud rate (~16%) despite lower volume than mobile.



Timing: fraud spikes in the 4–7am window across every day of the week.

Engineering decisions that make it robust



Separate identifiers from features

IDs (customer, device, IP) are kept aside, not encoded — preventing a 25,000-column blow-up, data leakage, and overfitting to past fraudsters.



Behaviour features, not identities

Velocity, transaction and device counts become the model's strongest, generalisable fraud signals (confirmed in the data).



Imbalance handled honestly

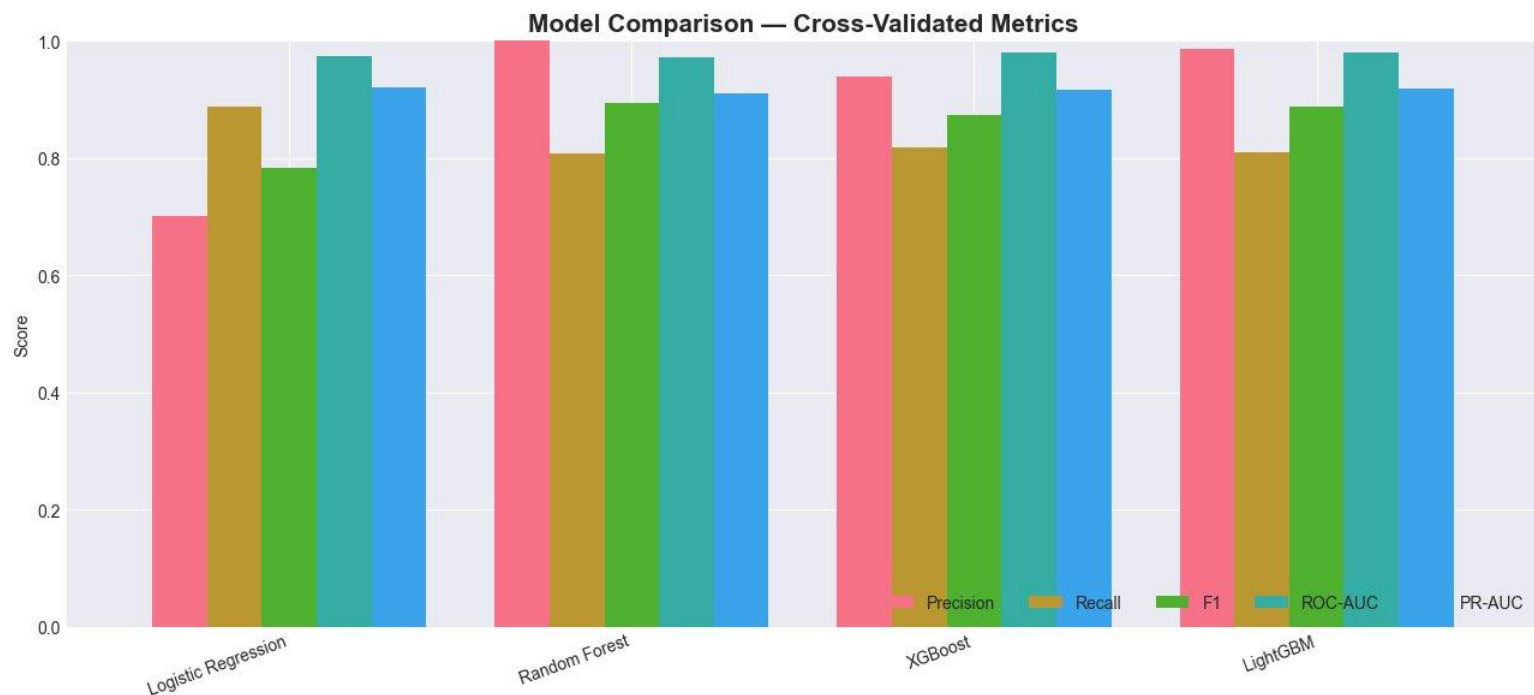
Stratified splits + class weighting; judged on recall, precision and PR-AUC — never on misleading accuracy.



Explainability built in

SHAP gives transaction-level reasoning, so analysts and regulators see why each transaction is flagged.

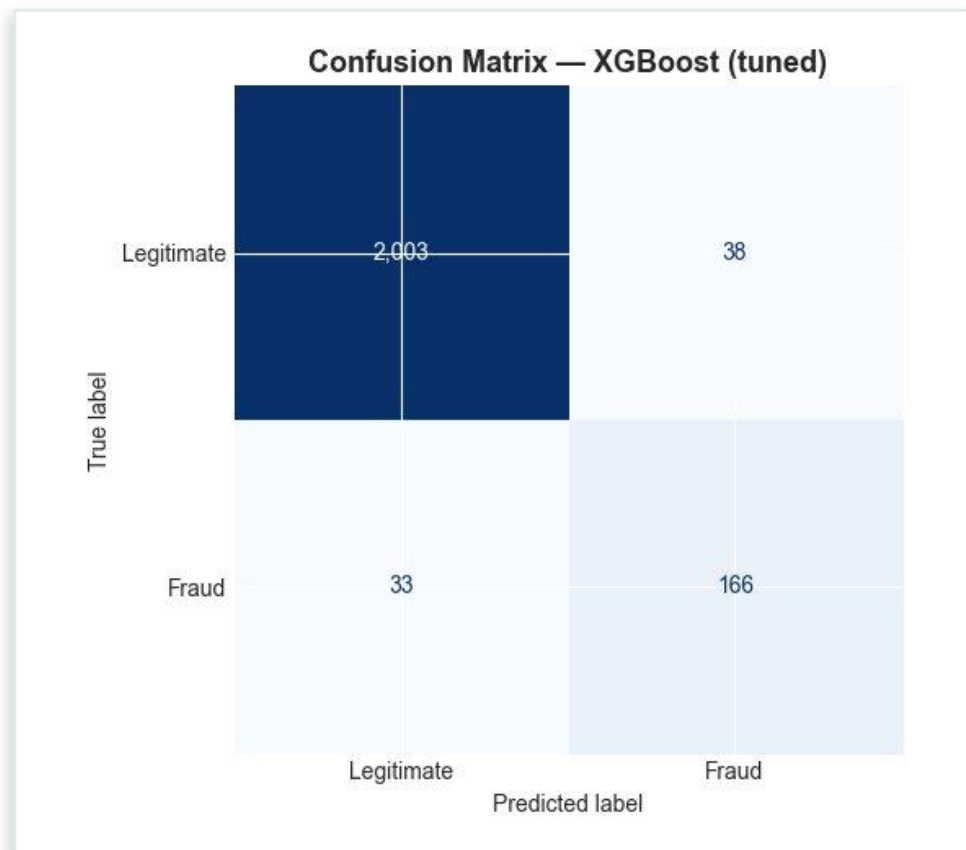
Four models compared — gradient boosting wins



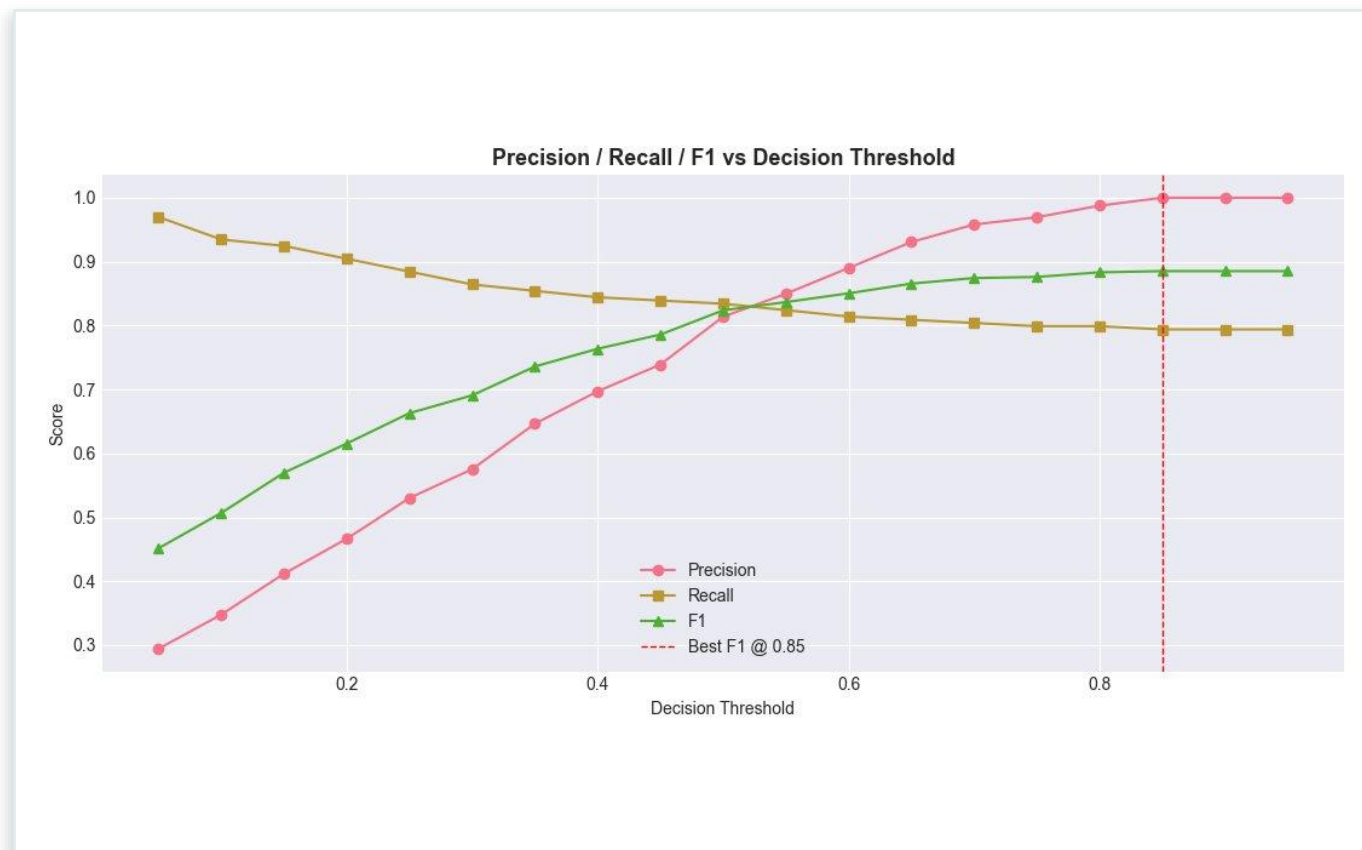
Reading it

- All four score high ROC-AUC (~0.97–0.98).
- XGBoost & LightGBM give the best precision–recall balance under imbalance (PR-AUC ~0.92).
- XGBoost (tuned) selected as the champion for deployment.

Champion performance & the business dial

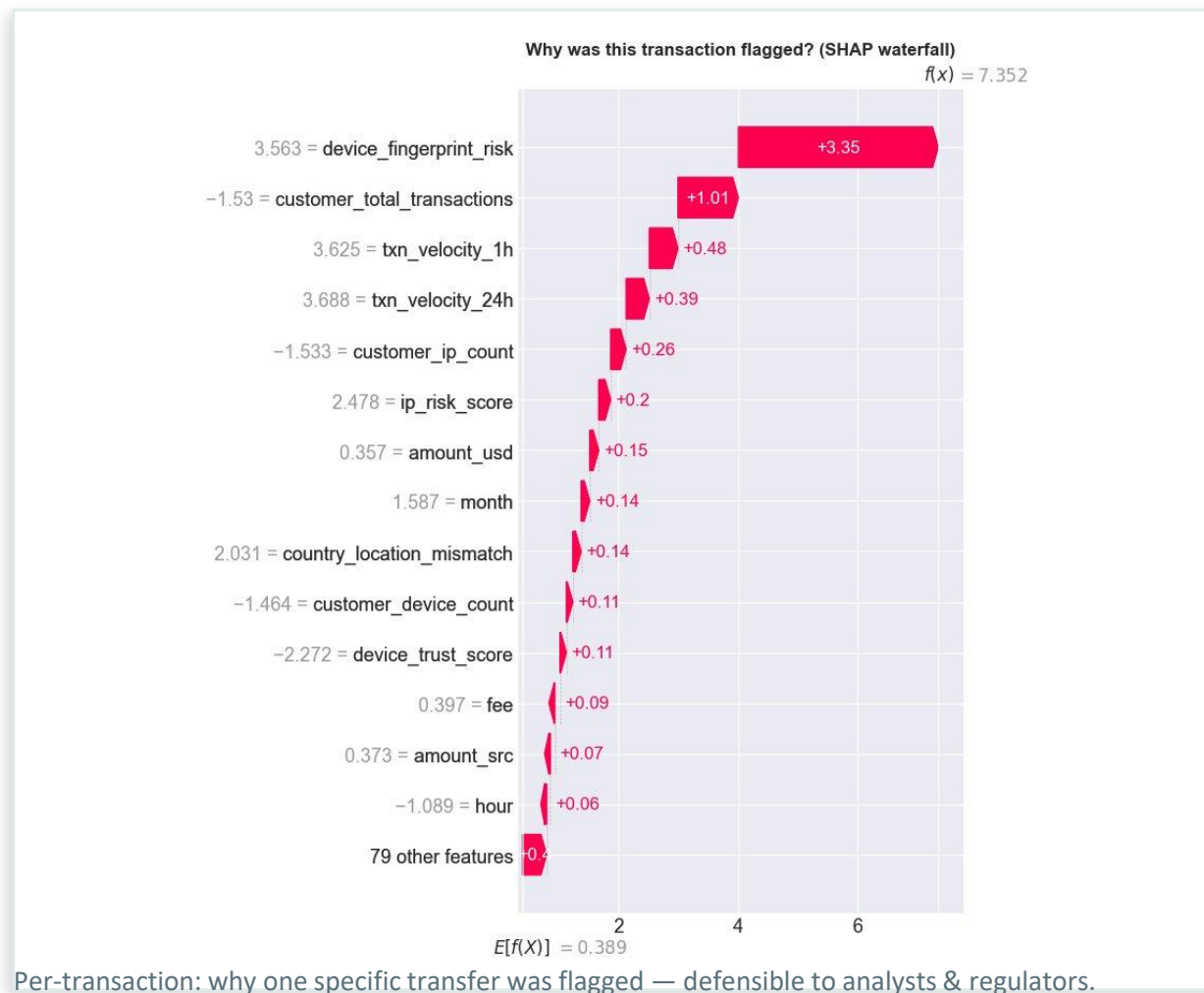
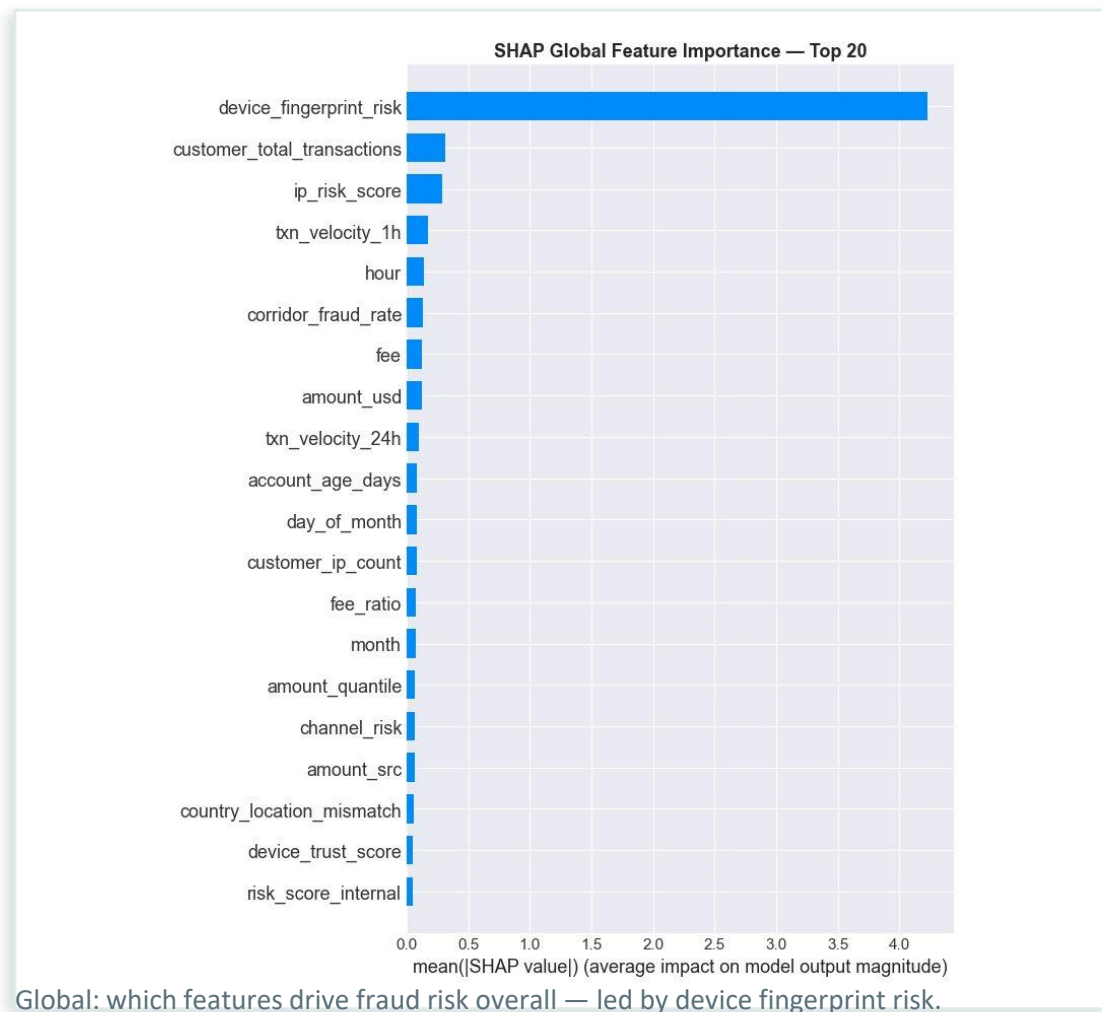


Confusion matrix: 166 frauds caught, 33 missed, only 38 false alarms.



Threshold = the business dial: slide up for fewer false alarms, down to catch more fraud. F1 optimised near 0.85.

Every decision is explainable (SHAP)



Robust, scalable, and efficient by design



Robust

- Cross-validated model selection
- Untouched holdout: 83% recall, 81% precision
- Defensive checks at every stage



Scalable

- Containerised FastAPI scoring
- Boosted models score in milliseconds
- Compact feature set — no ID explosion



Efficient

- 1.9% false-alarm rate → low analyst load
- Risk-ranked queue prioritises threats
- Automated retraining cuts manual upkeep

Cost to build, deploy, and run

One-off build investment, then a modest recurring run cost. Figures are indicative planning ranges.

Phase	Activity	Cost type
Data & modelling	Cleaning, EDA, model dev & tuning (complete)	One-off
Validation	Holdout testing, SHAP, sign-off (complete)	One-off
Deployment	FastAPI service, Docker, CI/CD, integration	One-off
Infrastructure	Compute & hosting for scoring API	Recurring
Monitoring	Drift detection, dashboards, alerting	Recurring
Retraining	Quarterly refresh as fraud evolves	Recurring



Build (one-off)

Front-loaded; data, modelling & validation are done



Run (recurring)

Hosting + monitoring + quarterly retraining — predictable

Cost offset: at 83% recall and a 1.9% false-alarm rate, every prevented fraud and avoided false block returns recovered losses and analyst hours.

Short, medium & long-term value



Short term

0–3 months

- Deploy scoring API alongside current rules
- Immediate lift: ~83% fraud caught at 1.9% false alarms
- Risk-ranked queue for analysts



Medium term

3–12 months

- Retire brittle rules as ML proves out
- Threshold tuned to business risk appetite
- Drift monitoring stabilises performance



Long term

12+ months

- Self-renewing model via quarterly retraining
- Extensible to new products & corridors
- Compounding data advantage over fraudsters

Recommendations & next steps



Approve shadow deployment

Run ML in parallel with current rules to confirm the 83% / 1.9% result on live traffic at zero customer risk.



Agree the operating threshold

Set the fraud-caught vs. false-alarm trade-off with Risk & Ops; F1-optimal sits near 0.85.



Stand up monitoring first

Drift detection and dashboards before cutover, so performance is observable from day one.



Invest in richer features

Graph/network signals and real-time velocity for the next accuracy step-change.



Close the analyst loop

Feed investigator outcomes back into retraining to sharpen the model continuously.



Plan multi-product rollout

Extend the same pipeline to new corridors and products once validated.



A smarter line of defence

83% of fraud caught · 1.9% false alarms · every decision explained. Adaptive where rules are static.

Recommended decision: approve shadow deployment and monitoring setup to confirm uplift on live traffic.